

INTERNATIONAL CONFERENCE ON EMERGING TRENDS IN NANOMATERIALS AND DEVICES (ICONMD-23)

2nd & 3rd March -2023

Organized by
Department of Physics

Vellalar College for Women (Autonomous)

College with Potential for Excellence

(Re-accredited with 'A' Grade (IV Cycle) by NAAC & Affiliated to Bharathiar University, Coimbatore - 641 046)
Erode, Tamilnadu, India.

ICONMD
2nd & 3rd March - 2023

Abstracts



MD39	Biological Activity and Characterization of Nickel oxide nanoparticles using Abutilon indicum leaf extract	K.G. Aarthe, T. Dharani
MD40	Biological and Photocatalytic Application of Copper oxide Nanoparticles using Dabur honey and Horn honey	K. Sowmyalakshmi, V. Kavithashri
MD41	Green Mediated Synthesis of Zinc oxide and Silver doped zinc oxide nanoparticles using Morinda citrifolia fruit extract	P. Sri Devi, V.S Anandhavarshini
MD42	Optical properties of Calcium oxide nanoparticles doped with Triton x-100(surfactant) using co-precipitation method	K.Sujatha, S. Vishnu Priya
MD43	Structural Morphology and Electrochemical behaviour of Nano structural MgCo_2O_4 for Supercapacitor applications	D.Iniyaraja, S.Deepika, T.Jothilakshmi, S.M.Senthilkumar N.Sivakumar.
MD44	Role of TiO_2 on Co_3O_4 positive electrode for energy storage applications	S.Deepika, T.Jothilakshmi, R. Kalaivani, N.Sivakumar
MD45	Synthesis, Characterization, antimicrobial, antioxidant and anticancer property of 6,6'-(2-methyl-1,3-phenylene)bis(azan-1-yl-1-ylidene)bis(methan-1-yl-1-ylidene)bis(2-methoxyphenol) ligand of Copper and Nickel complexes	R. Gandhimathi, S. Anbuselvi
MD46	Current-Voltage measurements of Au/Methylene Blue (MB)/n-Ge Schottky contact in a wide temperature range	M. Pavani, A. Ashok Kumar, V. Rajagopal Reddy, V. Janardhanam, Chel-JongChoi
MD47	Facile Synthesis of Porous Carbon Derived nis Nanocomposite for Long Cycle-Life Asymmetric Supercapacitors	M.Beemarao P.Periyannan N.Senguttuvan ^a , K.Ravichandran
MD48	Electrical Properties of ZnPc/n-Ge heterostructures	S.Guru Swathi, S.Ramesh, A.Ashok kumar, V.Rajagopal Reddy, V.Janardhanam, Chel-Jong Choi
MD49	Enhanced photodiode properties of Sn-doped LaPO_4 as an interfacial layer of Cu/Sn- LaPO_4 /n-Si type Schottky barrier diode.	R. Priya, R. Mariappan, R.N. Jayaprakash
MD50	A Review on Recent Advances in Synthesis and Applications of Conducting Polymer Nanocomposites	N. Dhachanamoorthis S.R. Gaaviyaa
MD51	Green Synthesis and Characterization of Magnesium oxide Nanoparticle using Psidium guajava	R.Boomathi, V. Kalaiselvi, P.Yasothea, B.Blessymol
MD52	Structural Analysis of Gallic acid ($\text{C}_6\text{H}_2(\text{OH})_3\text{CO}_2\text{H}$) through Density Functional Theory (DFT)	C. Priya, Sheeba Veronica, V. Kalaiselvi
MD53	Theoretical Studies of Benzoic Acid($\text{C}_6\text{H}_5\text{COOH}$) using Density Functional Theory(DFT)	C.Divyabharthi, Sheeba Vetonica, V.Kalaiselvi
MD54	Green Synthesis and Characterization of Mgo/Zno Nano composite used for Breast Cancer Applications	D.Divyashree, V.Kalaiselvi, B.Blessymol, P.Yasothea.
MD55	Green Synthesis of Titanium Dioxide TiO_2 Nanoparticles using Leucas aspera leaf extract	E.Hemapriya, B.Blessymol, P.Yasothea

MD48

Electrical Properties of ZnPc/n-Geheterostructures

S.Guru Swathi^{1,*}, S.Ramesh^{1,*}, A.Ashok kumar², V.Rajagopal Reddy³, V.Janardhanam⁴, Chel-Jong Choi⁴

¹Department of Physics,GSS,Gitam deemed to be University,Vishakhapatnam,530045

²Department of Physics,YSR Engineering college of YVU,Proddatur,A.P,India

³Department of Physics, Sri Venkateswara University, Tirupati,A.P, India.

⁴Semiconductor Physics Research Center(SPRC),School of Semiconductor and Chemical Engineering, Jeonbuk National University,Jeonju 54896, Korea.

Abstract

The electrical properties of Au/n-GeSchottky contacts with Zinc Pthalocyanine (ZnPc) interfacial layer is analysed using current-voltage (I-V) properties. The Schottky barrier parameters such as barrier height (Φ_b), ideality factor (n), was measured for the contacts using the I-V characteristics at low forward bias voltages ($V < 3kT/q$) based on thermionic emission model. The barrier height and ideality factor of the Au/n-Ge contact shows 0.57 eV and 1.29 and 0.72 eV and 1.21 for Au/ZnPc/n-Ge contacts respectively. Series resistance of the contacts was also evaluated using Cheung's and Norde method. A high series resistance of 718Ω is observed in $dV/d\ln I$ versus I plot for Au/ZnPc/n-Ge contacts than compared to 280Ω in the case of Au/n-Ge contacts. Large increase of shunt resistance (R_{sh}) from $17.27 K\Omega$ (Au/n-Ge) to $3.89 M\Omega$ (Au/ZnPc/n-Ge) is observed. The current conduction mechanisms of the contacts were analysed using logarithmic I-V plot. Three distinct linear regions were observed for Au/n-Ge contacts and the slope factor ($m \sim 2$) of these linear regions confirms the ohmic conduction mechanism. In the case of Au/ZnPc/n-Geheterostructures, four distinct regions (region-I to region IV) are observed and the slope factor of region-I ($m \sim 2$) shows ohmic conduction mechanism and space charge limited current (SCLC) is observed in region-II ($m > 2$). The slope factor of region III ($m \sim 4$) signifies the current conduction mechanism is due to Trap charge limited current conduction mechanism. Finally, the current conduction mechanism in region-IV ($m > 2$) indicates trap free SCLC mechanism. Analysis of electrical properties of ZnPc/n-Geheterostructures is very essential to use these devices in different applications like organic light emitting diode, photo-generated charges, organic field effect transistors, gas sensors, and especially in photovoltaic devices.

Keywords: n-Ge, Schottky contacts, Heterostructures, ZnPc, Electrical properties, Current conduction mechanism

Corresponding Author Name: S.Guru Swathi¹

E Mail: sguru@gitam.in

VELLALAR GROUP OF INSTITUTIONS

Vellalar College for Women (Autonomous)
Vellalar Matriculation Higher Secondary School for Girls
Vellalar College of Engineering and Technology
Vellalar College of Education for Women
Vellalar High School for Girls (State Board)
Vellalar Teacher Training Institute (Co-education)
Vellalar College of Nursing
Vellalar Vidyalayaa Senior Secondary School (CBSE)
Vellalar College of Pharmacy (B.Pharm & D.Pharm)
VET Institute of Arts and Science (Co-education)
Vellalar Educational Trust Charity Hospital



ROYAL BOOK PUBLISHING

Address :
52-1 Ward No : 10 K.M. NAGAR,
Ayothiyapattinam, Salem - 636 103.
Tamilnadu, India.
E-mail : contact@royalbookpublishing.com
Website : www.royalbookpublishing.com

